Dev Singhal

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TECHNICAL SKILLS

Computer Vision & ML: YOLO, CNNs, PyTorch, OpenCV, Multi-View Geometry, Object Detection

Programming & Tools: Python, C++, Java, Docker

Database & Backend: MySQL, Firebase

Additional: Flutter, Twilio API, ESP32, Sensors

EDUCATION

B.Tech in Computer Science | Amity University, Noida | 2021-2025 | 8.26 CGPA

WORK EXPERIENCE

Full Stack Developer ***June 2024 – August 2024***
Ulavi Technologies Pvt. Ltd., Singapore (Remote)

- Developed WhatsApp chatbot using LLM with **MySQL database** for persistent data storage and query optimization
- Integrated Twilio API and implemented efficient backend using Python, reducing response time by 40%

PROJECTS

Multi-View Geometry Object Detection System ***| Awarded "Best Startup Potential" - Technovate 2025***

- Developed multi-camera object detection system using **YOLO** achieving 95%+ real-time tracking accuracy
- Implemented **CNNs** and multi-view geometry principles for 3D object position triangulation
- Trained and optimized models using **PyTorch** on custom datasets, reducing false positives by 30%

SunSense - Solar Panel Monitoring System ***| Hardware + ML***

- Built custom **CNN classification model** detecting panel contamination with 92% accuracy
- Stored sensor data and maintenance logs in **MySQL database**, improving panel efficiency by 18% on a self-hosted server on a raspberry pi
- Deployed model on ESP32 with automated cleaning triggers based on real-time predictions

Library Management System ***| Flutter + Firebase***

- Developed full-stack app managing 1000+ book transactions with automated fine calculations and alerts
- Implemented role-based access control for students and librarians with real-time database synchronization

Smart Home Automation ***| Hardware + IoT***

- Designed ESP32-based climate control system with Google Assistant integration and real-time sensor monitoring

ACHIEVEMENTS

- **Technovate 2025:** Won "Best Startup Potential" award, judged by DRDO engineers for CV-based project
- **Smart India Hackathon 2023 Finalist:** Led team of 5 building fitness tracking app with ML-based personalized recommendations for MathWorks
- **International Conference IT Support:** Led team of 3 for US-India Higher Education Conference at FDU, New Jersey